

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458895.835 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.057
Transit depth (Rp/Rs)^2	3.31 +/- 2.08 [%]
Orbital Inclination (inc)	80.19 +/- 2.8
Airmass coefficient 1 (a1)	3.23 +/- 1.73
Airmass coefficient 2 (a2)	-0.8 +/- 1.18
Scatter in the residuals of the lightcurve fit is	54.89 %
Best Comparison Star	None
Optimal Aperture	1.82
Optimal Annulus	6.75
Transit Duration (day)	0.042 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.182 $\pm$ 0.057 vs 0.1594 (deviation 0.32 σ)
Duration fit vs published	0.042 d vs 0.0512 d (deviation 0.47 σ)
Mid-time O-C	-4.1 min
Depth SNR	2.6
Residual scatter	54.89 %

Light curve

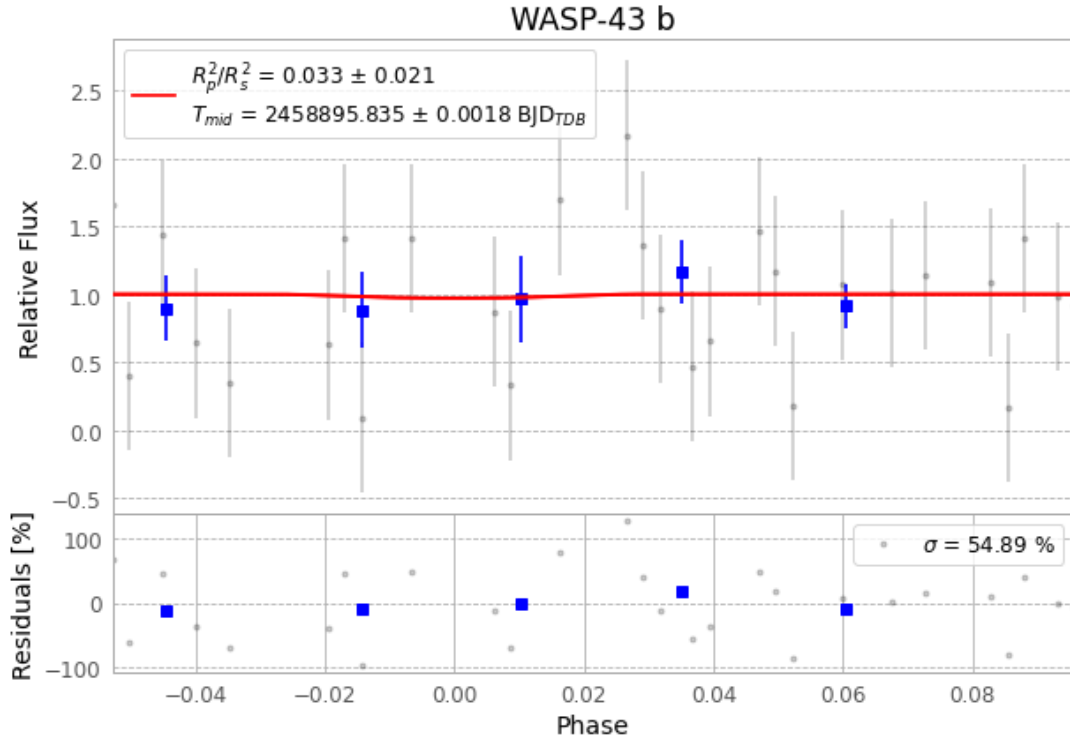


Figure 1 — FinalLightCurve_WASP-43 b_2020-02-15.png (SHA-256 072f529c3a37829c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [483, 207], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0a24b9fb101f5859...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), WASP-43200216063612.FITS → WASP-43200216094512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-200216.log (d483571159a4...), FinalLightCurve_WASP-43 b_2020-02-15.png (072f529c3a37...), FinalLightCurve_WASP-43 b_2020-02-15.csv (ef0c0d77ebe1...)

Compute resources

Wall time 112.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 23:50Z.